universidade de aveiro

KonetiQ

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OUR GOAL

Develop a platform to monitor and manage corporate micromobility fleets alongside personal vehicle owners.



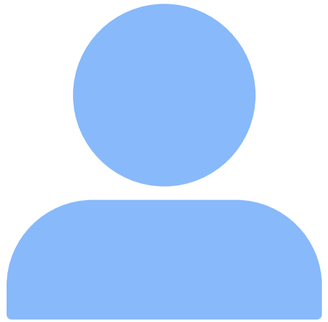
STATE OF THE ART

	Real-time Telemetry	Analytics / Business intelligence	ML-based module for anomaly detection	Vehicle lifecycle	Rider component	Open and extensible API	CO2 footprint tracking
Joyride	✓	✓	✗	✓	✗	✓	✗
Levy Fleets	✓	✓	✗	✓	✗	✓	✗
Comodule	✓	✓	✗	✗	✓	✓	✓
Trackap	✓	✓	✗	✓	✓	✗	✓
KonetiQ	✓	✓	✓	✓	✓	✓	✓

STATE OF THE ART

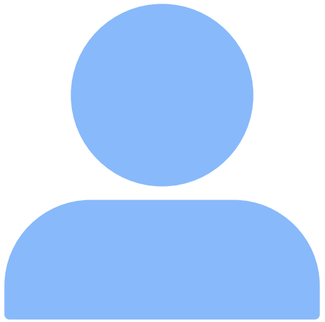
	Mobile App	Web App	Backend Core	Message Broker	Infrastructure
Joyride					
Levy Fleets					
Comodule					
Trackap					

ACTORS AND USE CASES



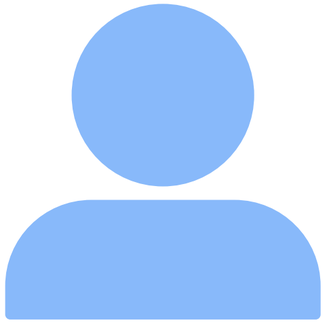
Fleet Operator

Manages users, modules and daily fleet operations, tracks live telemetry streams, controls asset lifecycle, and configures organization modules.



Maintenance Technician

Receives automated fault alerts, consults component service history, and executes physical repairs and operational state updates.



Fleet Analyst

Evaluates aggregated historical data via BI dashboards, inspects ML demand forecasts, and audits fleet-wide ESG CO_2 savings.



Rider

Uses the mobile companion app to monitor real-time ride telemetry, review personal journey history, view bike maintenance status and track individual eco-impact.

USE CASE DIAGRAM



FUNCTIONAL REQUIREMENTS

1

The system must continuously **ingest high-frequency telemetry data streams** emitted by vehicles

2

The system must **distribute validated telemetry events** in **real time** to connected web and mobile applications using WebSockets.

3

The system must **manage the complete lifecycle** of each vehicle using a **finite state machine** (Available, In-Use, In-Maintenance, Retired) to enforce legitimate operational transitions and maintain full service traceability.

4

The system must **analyze streaming and historical telemetry** via Machine Learning models to **identify component degradation and operational anomalies** for predictive maintenance.

5

The system must **compute carbon emission savings per ride and fleet-wide** by benchmarking vehicle energy consumption against equivalent fossil-fuel transport baselines.

6

The system must **expose documented, extensible REST/event APIs** to **decouple core fleet operations** from downstream analytics, Business Intelligence (BI), and predictive modules.

7

The web interface must **render an interactive map** displaying live vehicle locations, battery levels, operational statuses, and aggregated fleet/ESG KPIs.

8

The system must **authenticate users** and **enforce fine-grained permissions** across defined platform roles.

NON-FUNCTIONAL REQUIREMENTS

1

The end-to-end latency from telemetry packet ingestion at the broker to temporal persistence and WebSocket broadcast **must not exceed 500 ms.**

2

The data ingestion pipeline must **continuously process a sustained throughput of at least 500 messages/ second** without event drops while **maintaining latency thresholds.**

3

At least **95% of synchronous read requests** made to the REST API (such as bike status checks and KPI calculations) must **respond within 200 ms** under nominal workload.

4

All external network traffic across web, mobile, and IoT clients must be s**trictly encrypted** using HTTPS / TLS 1.3, WSS, and MQTTS, with **authentication and access tokens enforced** via Keycloak.

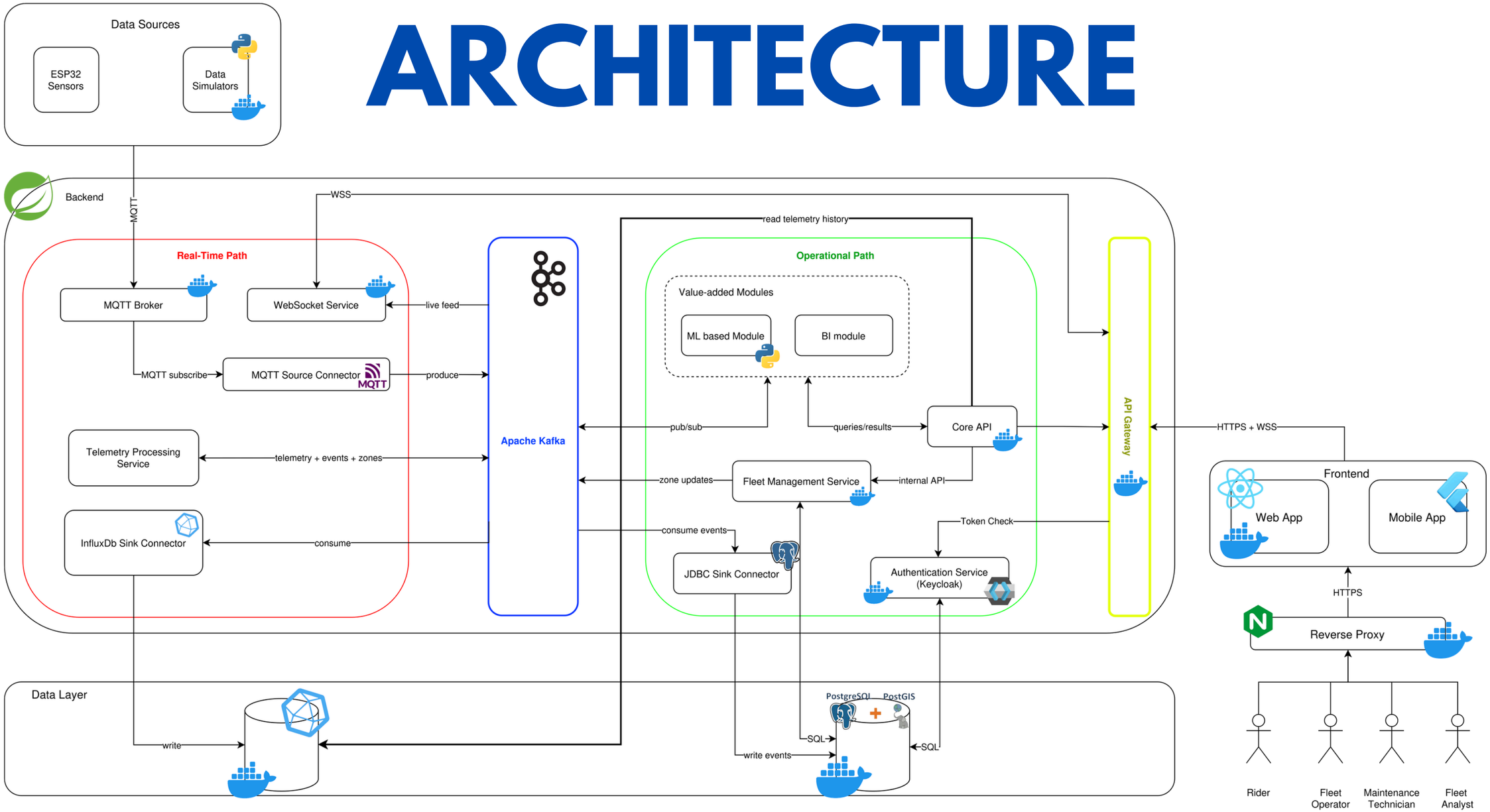
5

In the event of transient downstream database or service outages, the event broker (Kafka) must **durably retain telemetry messages** within configured retention policies to **ensure zero data loss upon recovery.**

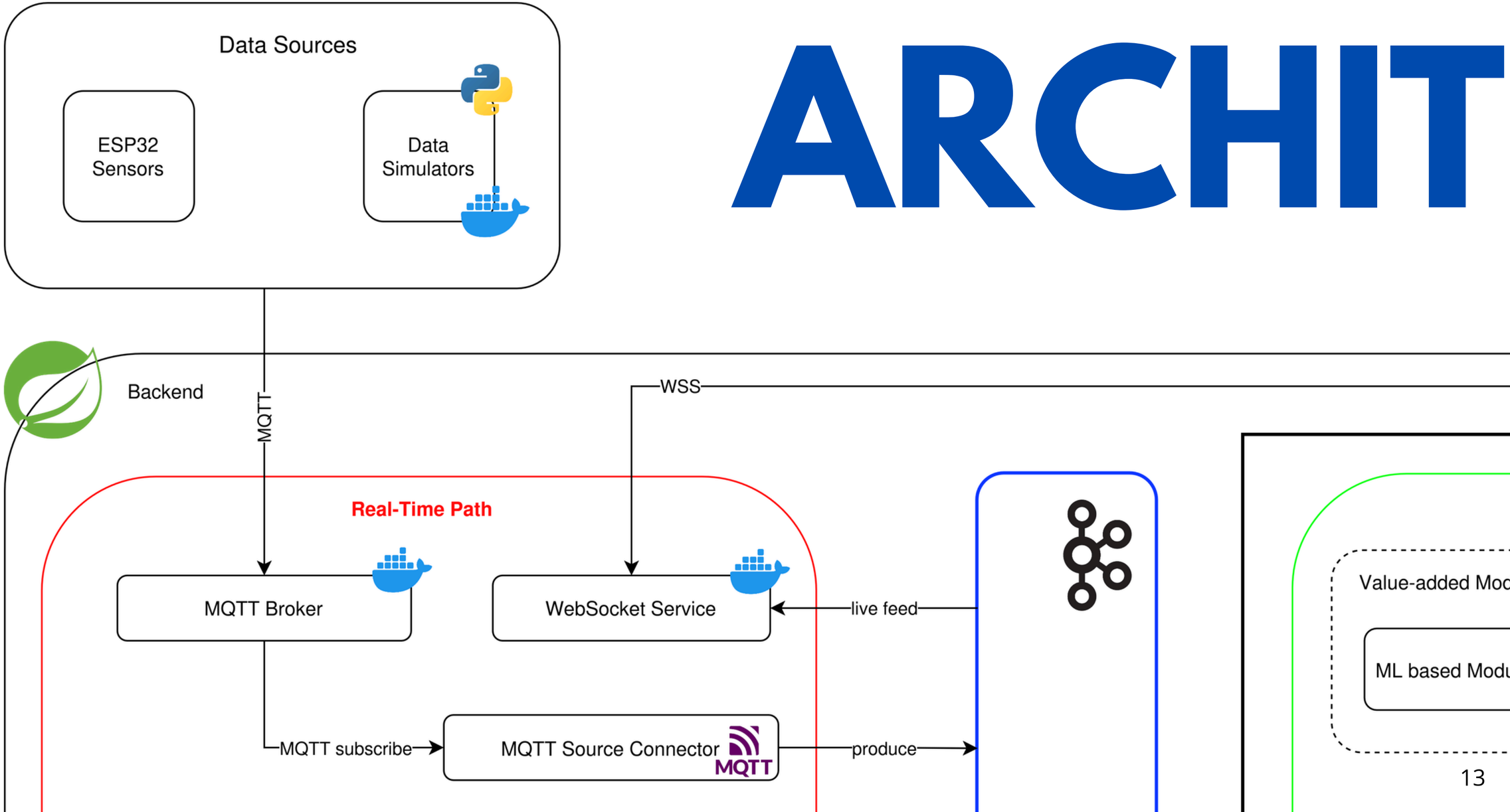
All backend services, stream brokers, connectors, and database stores must **be containerized** using Docker and orchestrated declaratively to facilitate **automated deployment and environment parity.**

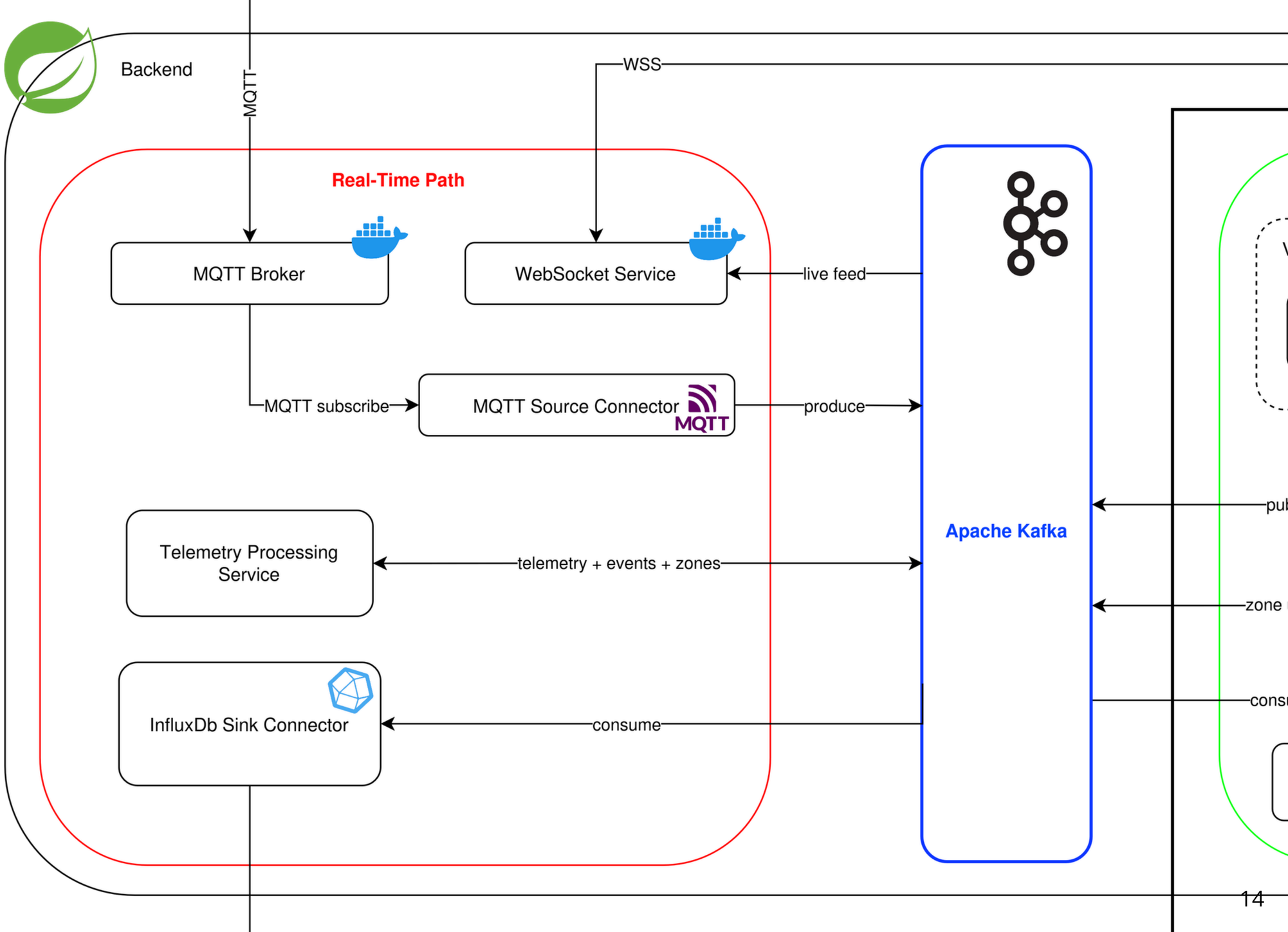
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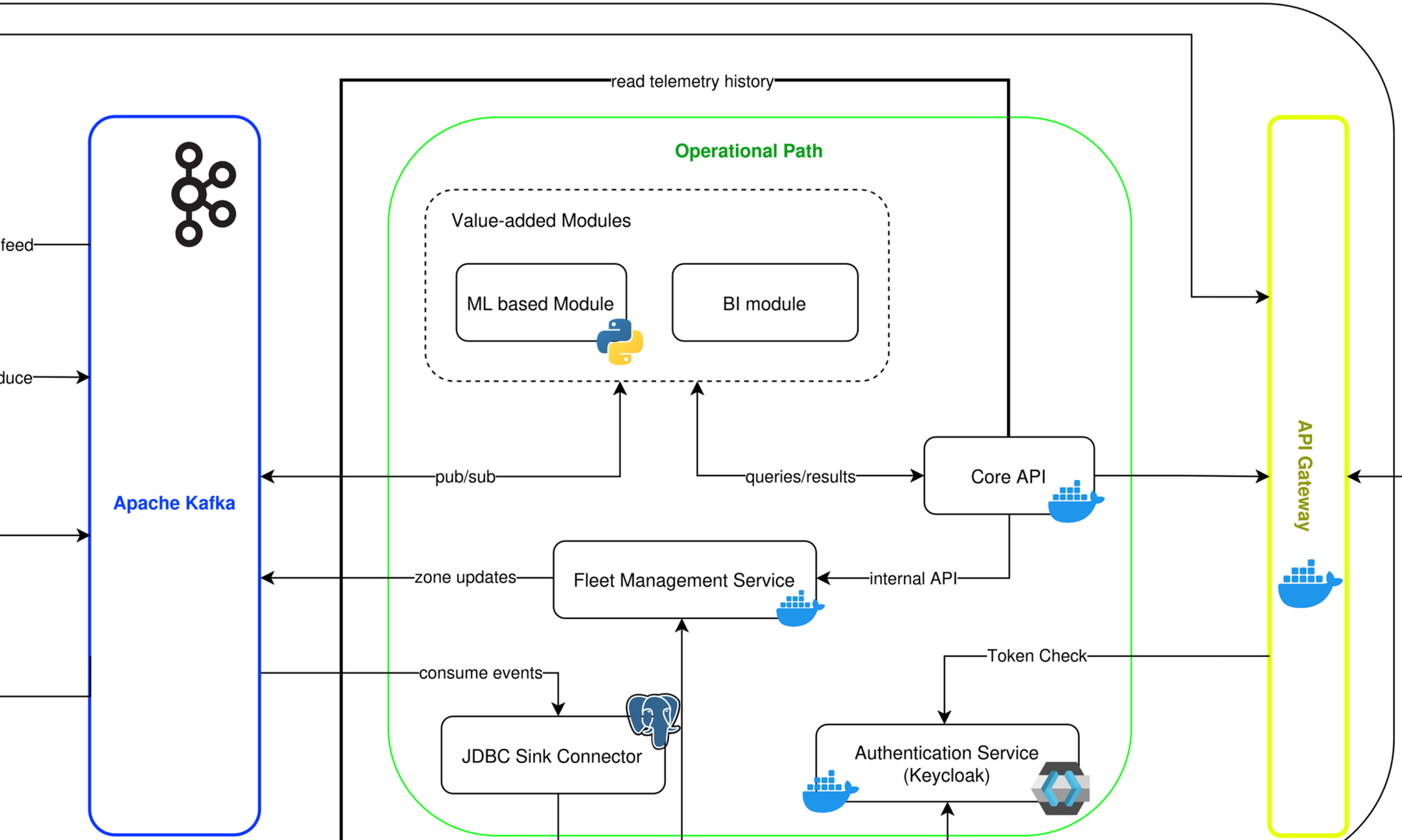
ARCHITECTURE

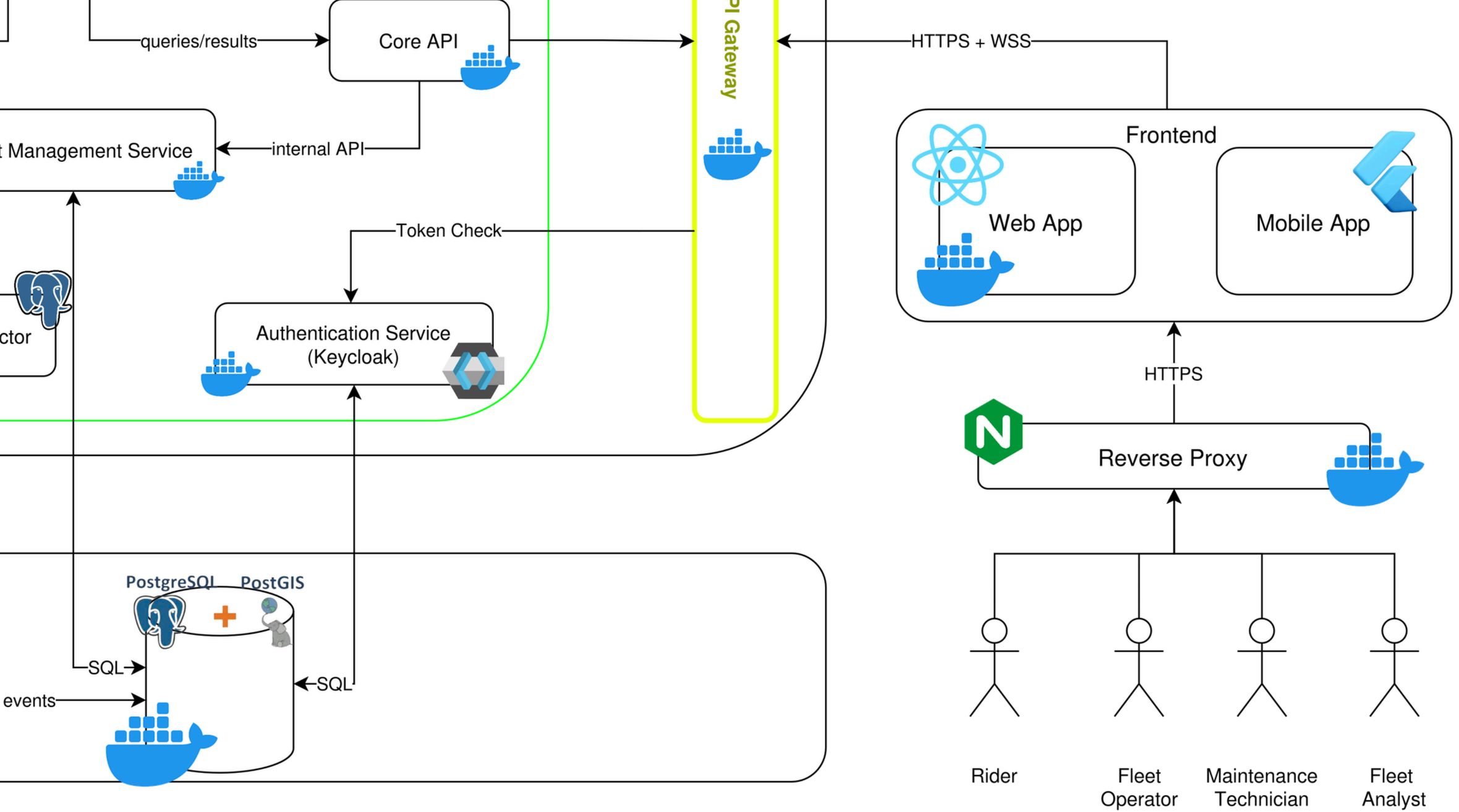


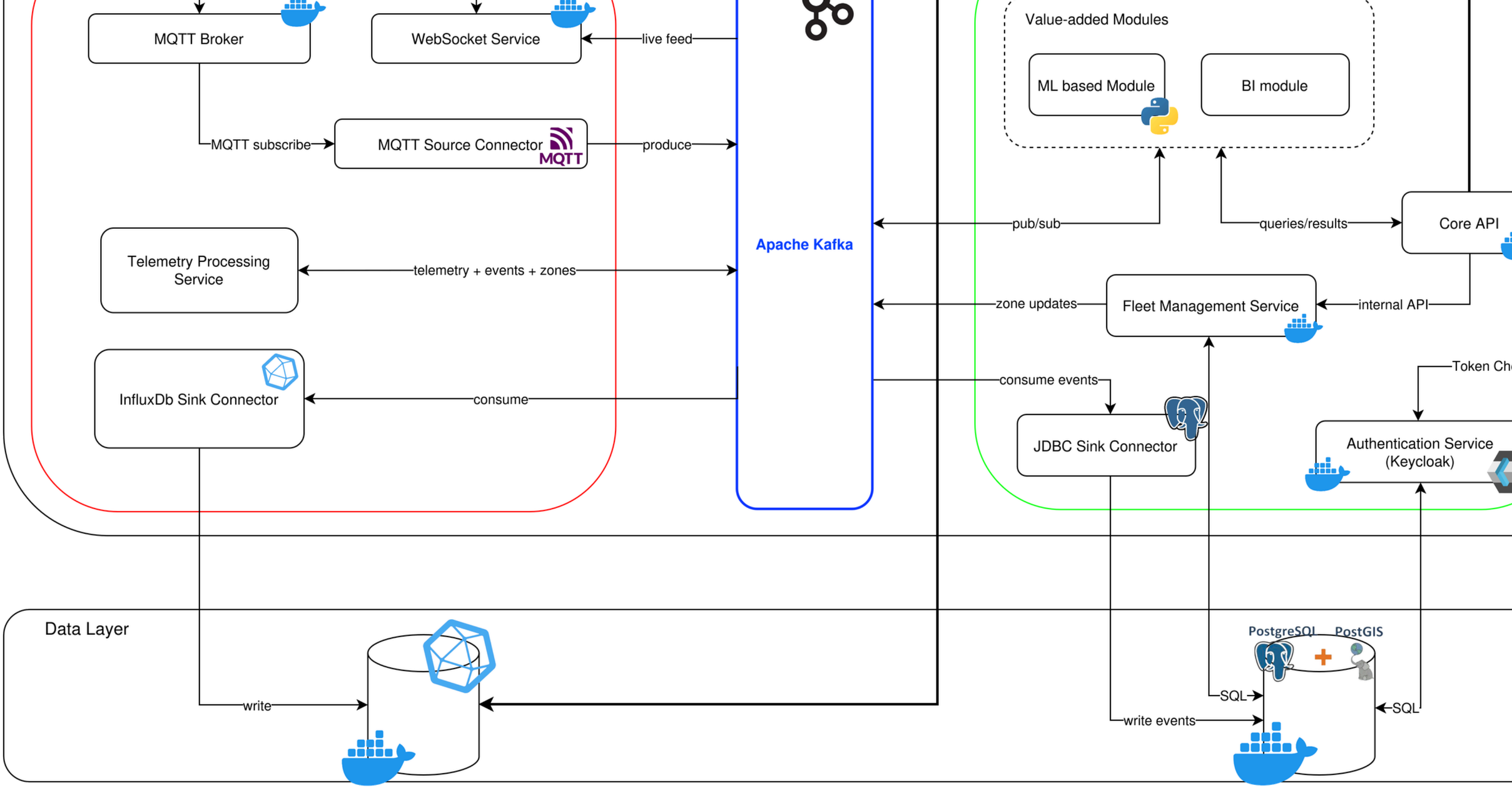
ARCHIT


Data Sources
ESP32 Sensors
Data Simulators
Backend
MQTT
WSS
Real-Time Path
MQTT Broker
WebSocket Service
live feed
MQTT subscribe
MQTT Source Connector
MQTT
produce
Value-added Mod
ML based Modu

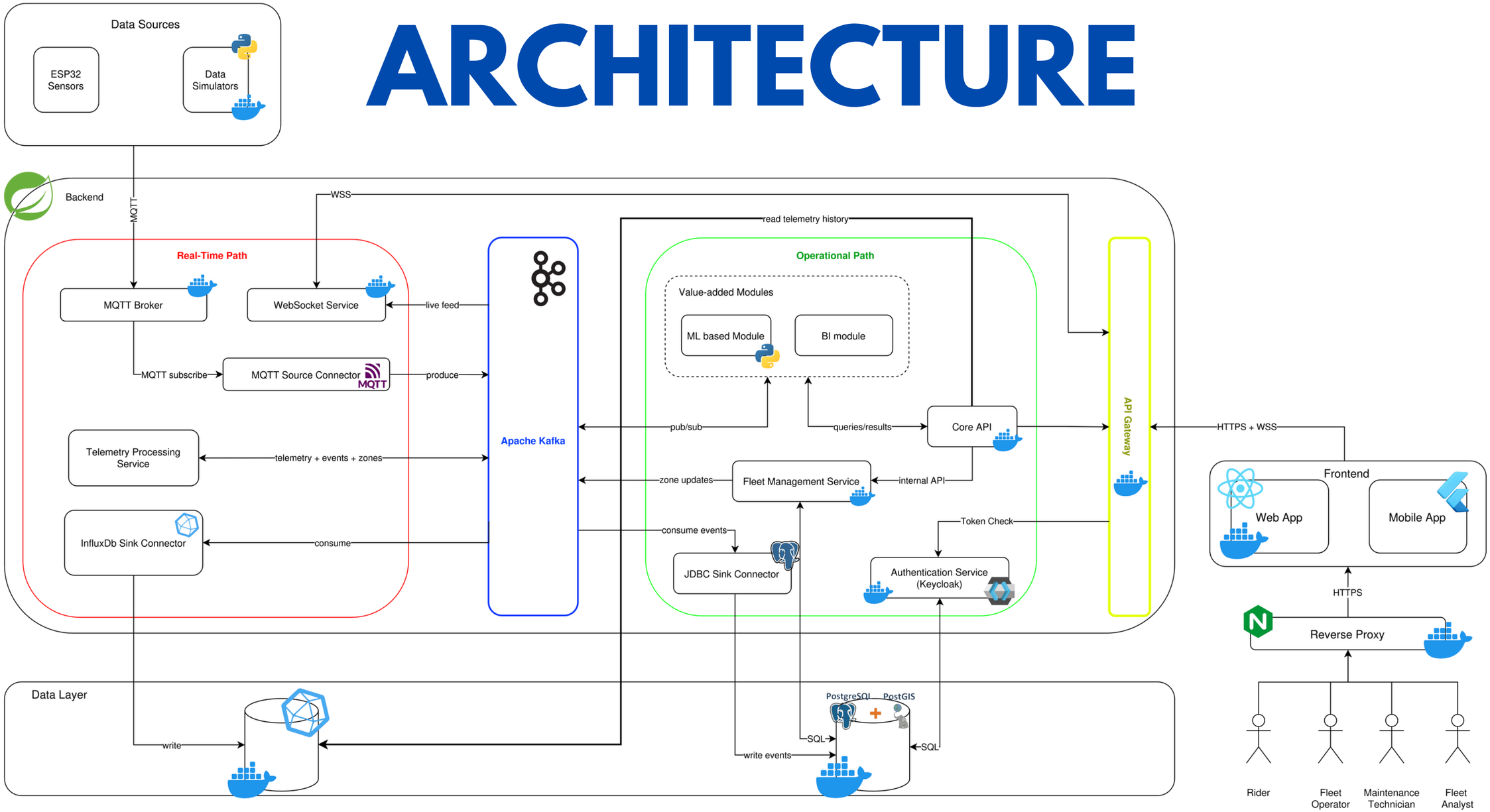

Backend
MQTT
WSS
Real-Time Path
MQTT Broker
WebSocket Service
live feed
MQTT subscribe
MQTT Source Connector
MQTT
produce
Apache Kafka
Telemetry Processing Service
telemetry + events + zones
InfluxDb Sink Connector
consume


read telemetry history
Operational Path
Value-added Modules
ML based Module
BI module
feed
duce
Apache Kafka
pub/sub
queries/results
Core API
API Gateway
HTTPS + WSS
zone updates
Fleet Management Service
internal API
consume events
Token Check
JDBC Sink Connector
Authentication Service
(Keycloak)
Web A

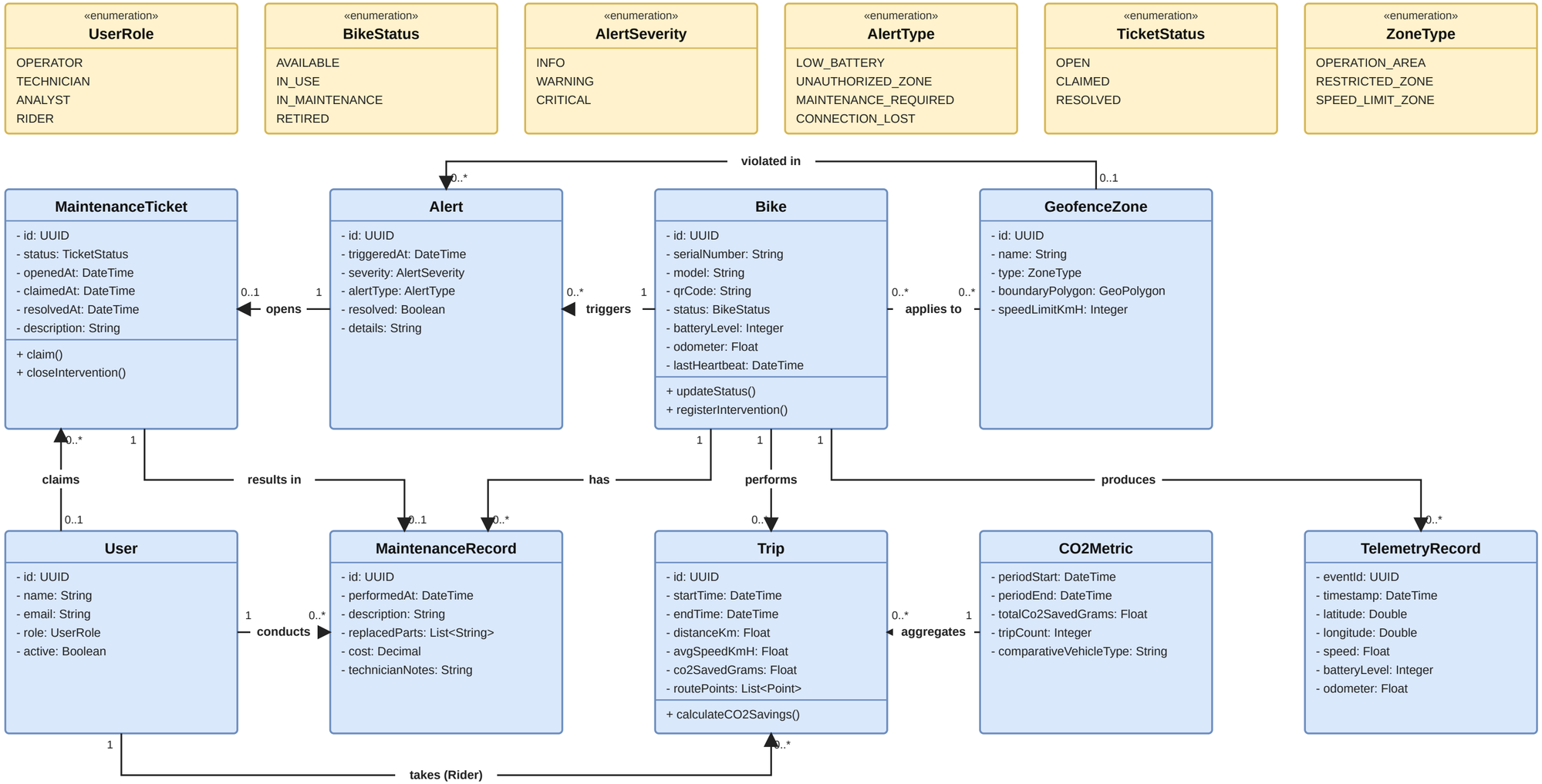

queries/results
Core API
t Management Service
internal API
Token Check
ctor
Authentication Service
(Keycloak)
PI Gateway
HTTPS + WSS
Frontend
Web App
Mobile App
HTTPS
Reverse Proxy
PostgreSQL
PostGIS
SQL
SQL
events
Rider
Fleet
Operator
Maintenance
Technician
Fleet
Analyst


MQTT Broker
WebSocket Service
live feed
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zone updates
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Token Ch
Authentication Service (Keycloak)
Data Layer
write
PostgreSQL
PostGIS
SQL
SQL
write events

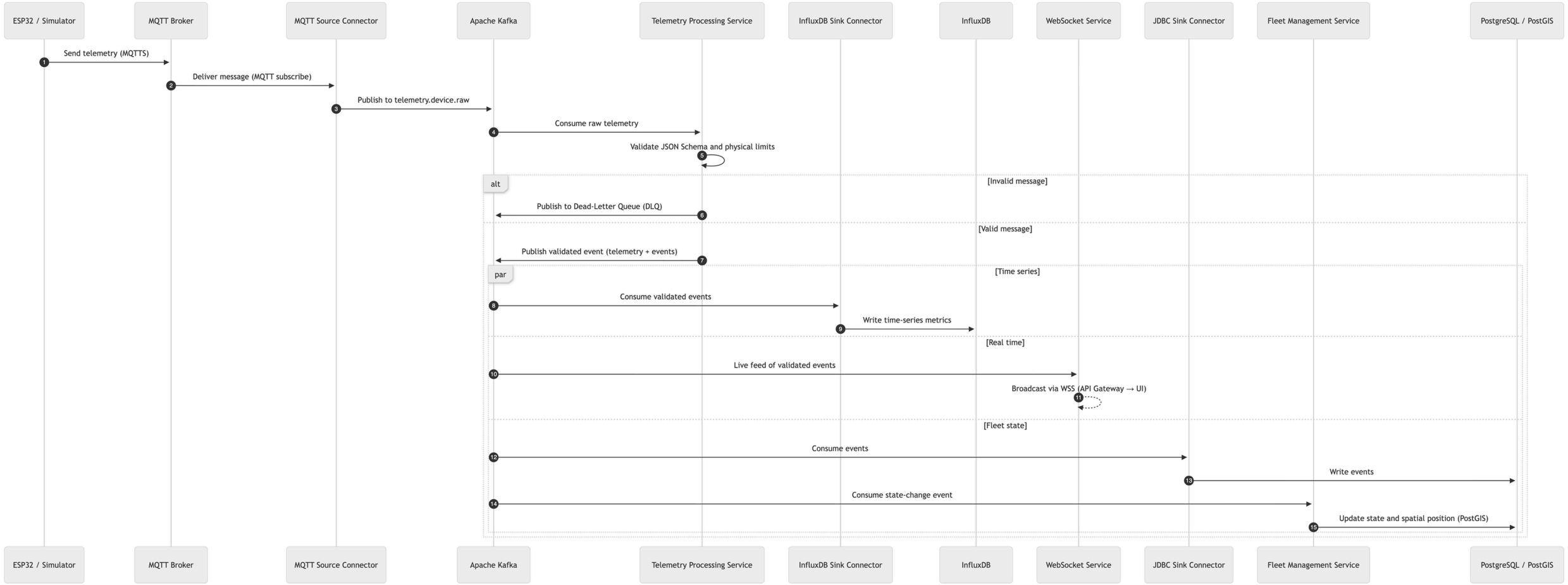
ARCHITECTURE



DOMAIN MODEL

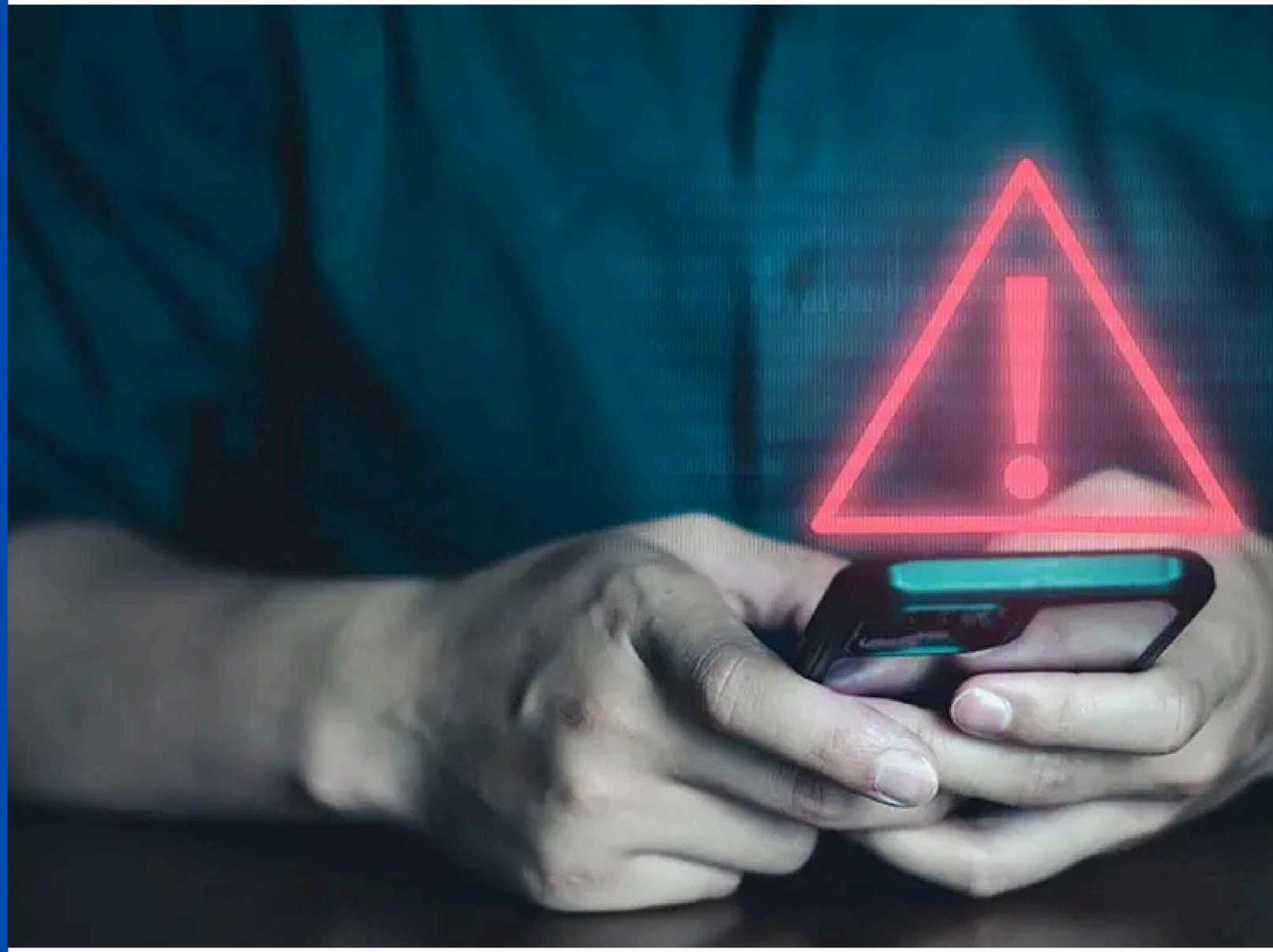


REAL-TIME TELEMETRY AND EVENT INGESTION FLOWS

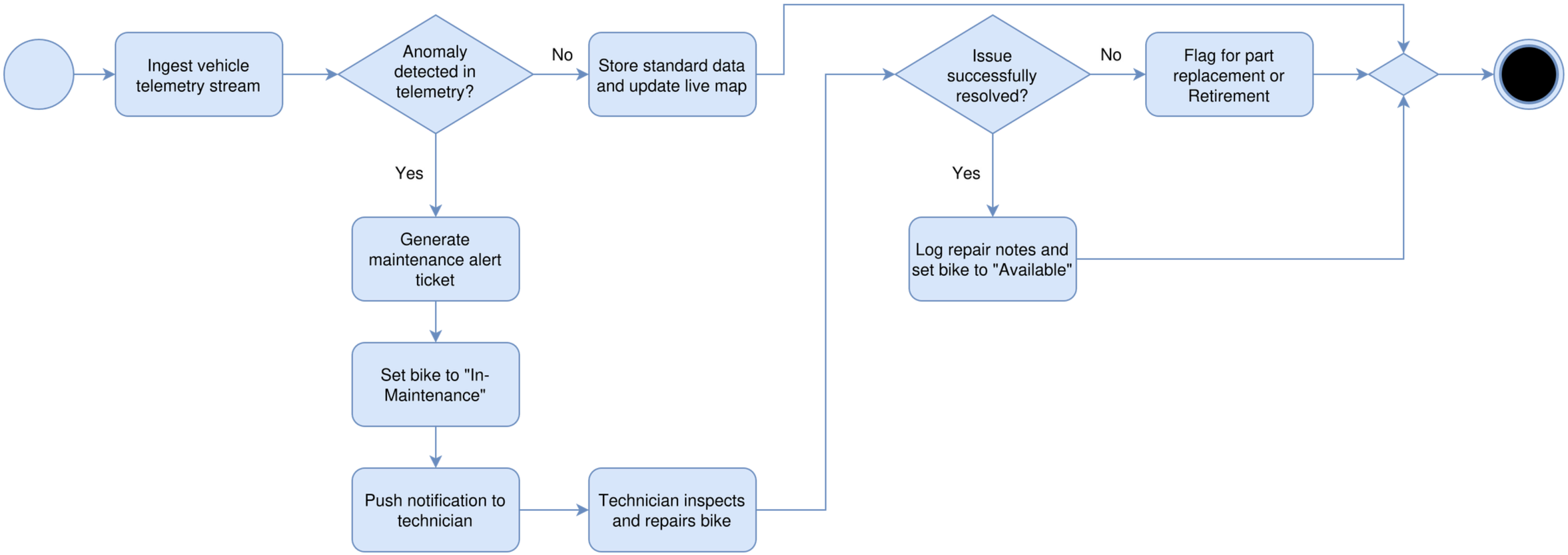


USE CASE DEEP-DIVE:

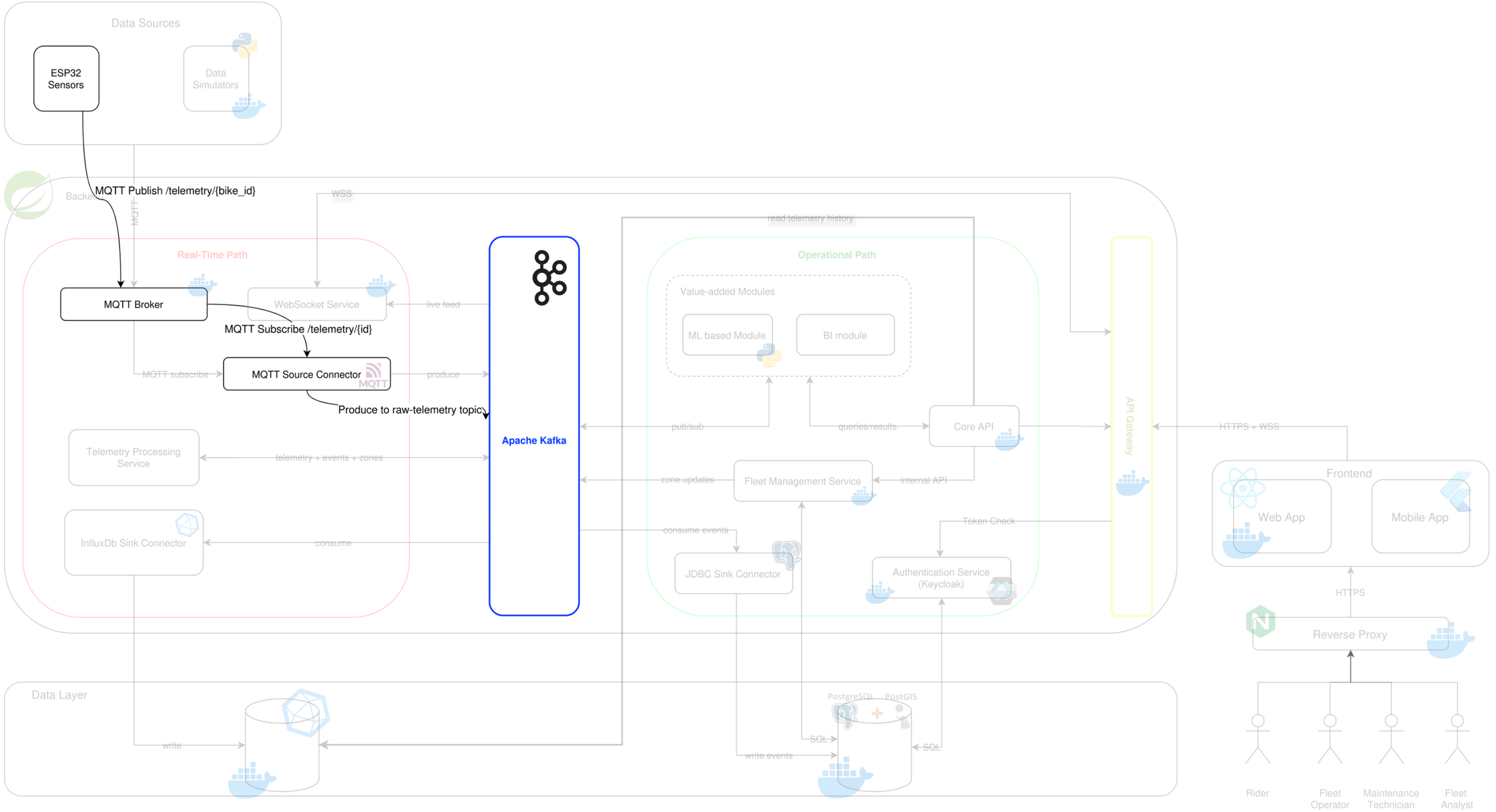
Receive maintenance alerts



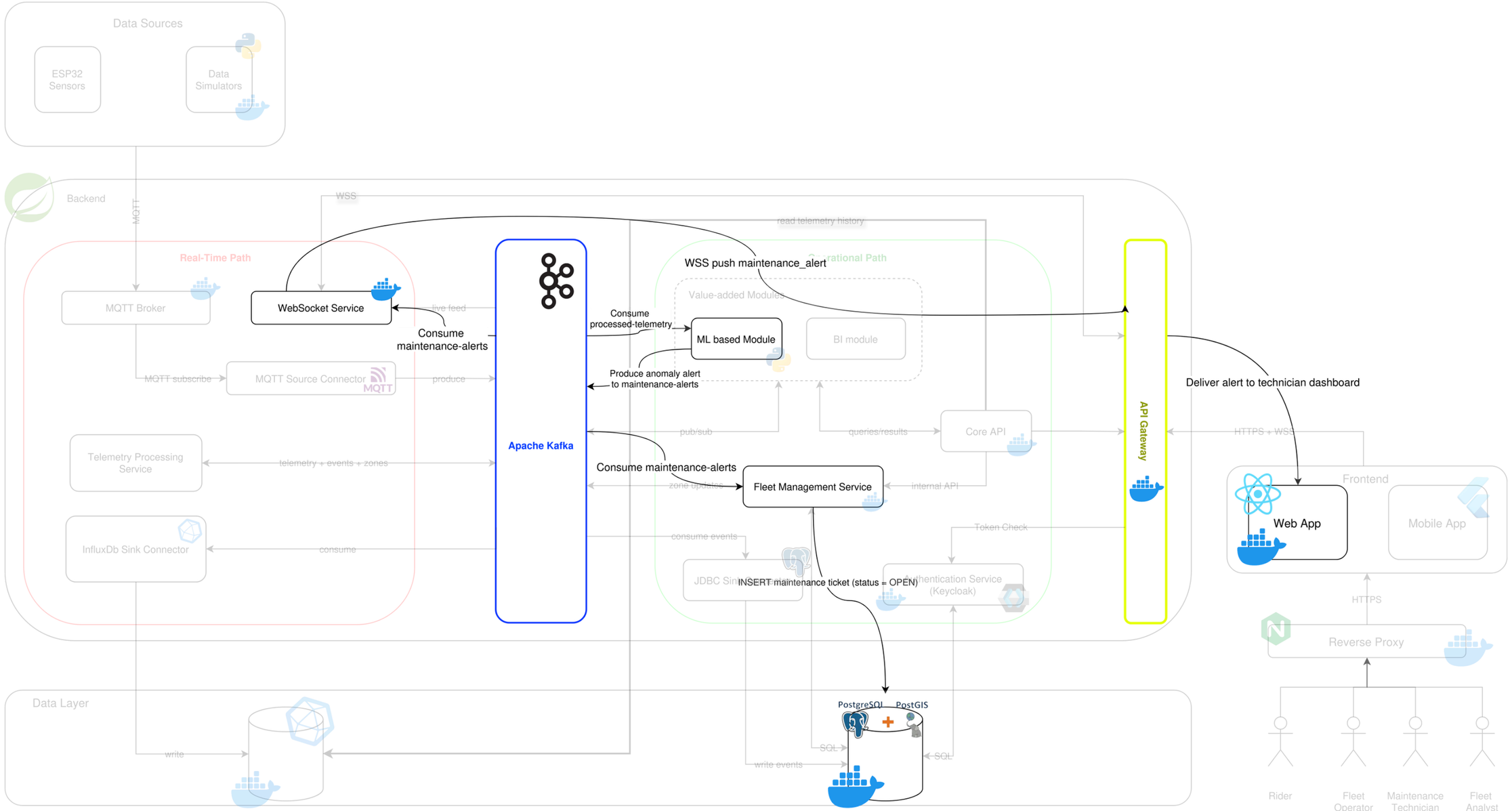
FLOWCHART


Ingest vehicle telemetry stream
Anomaly detected in telemetry?
No
Store standard data and update live map
Yes
Generate maintenance alert ticket
Set bike to "In-Maintenance"
Push notification to technician
Technician inspects and repairs bike
Issue successfully resolved?
No
Flag for part replacement or Retirement
Yes
Log repair notes and set bike to "Available"

ARCHITECTURE WORKFLOW

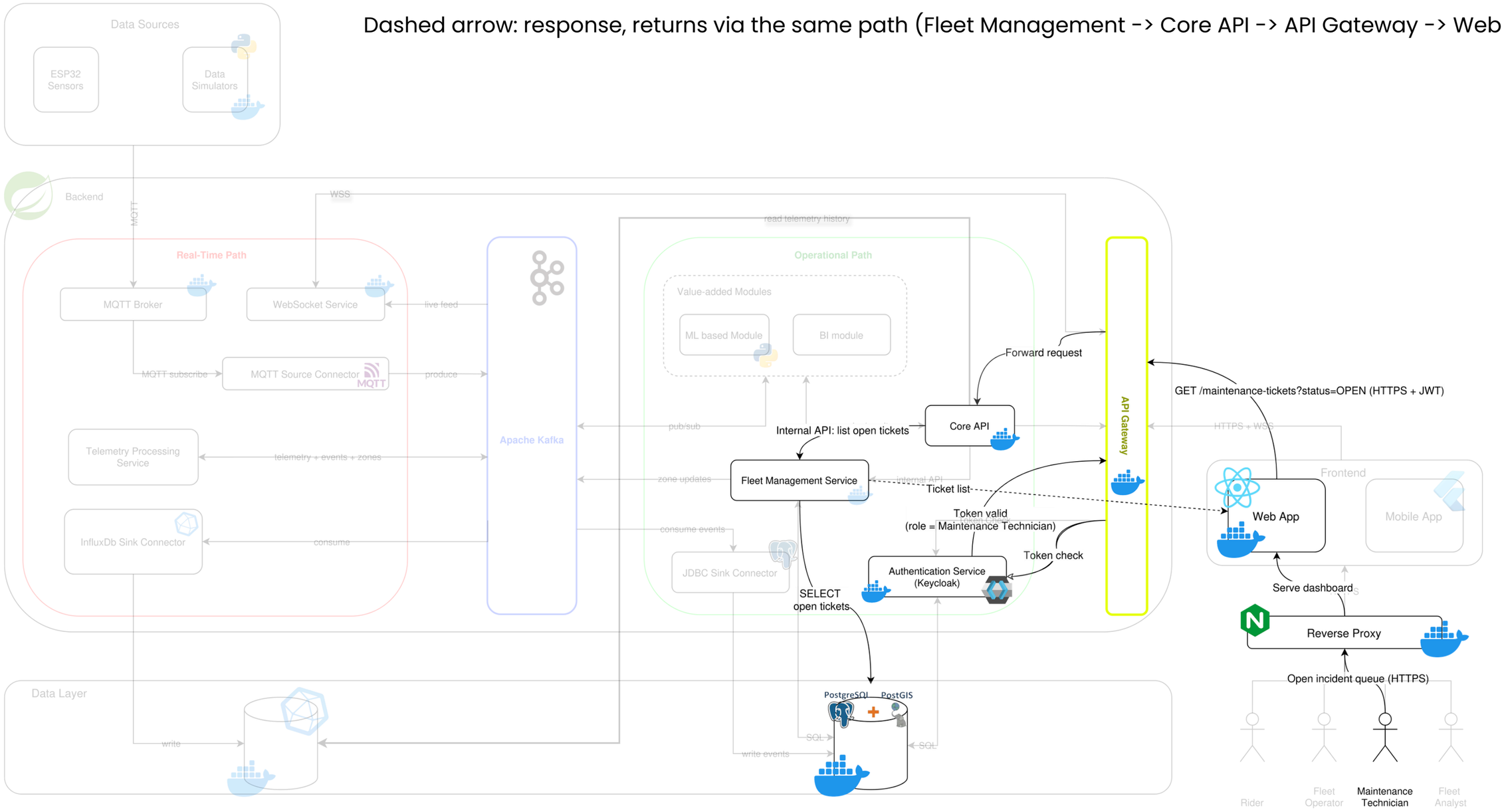


ARCHITECTURE WORKFLOW



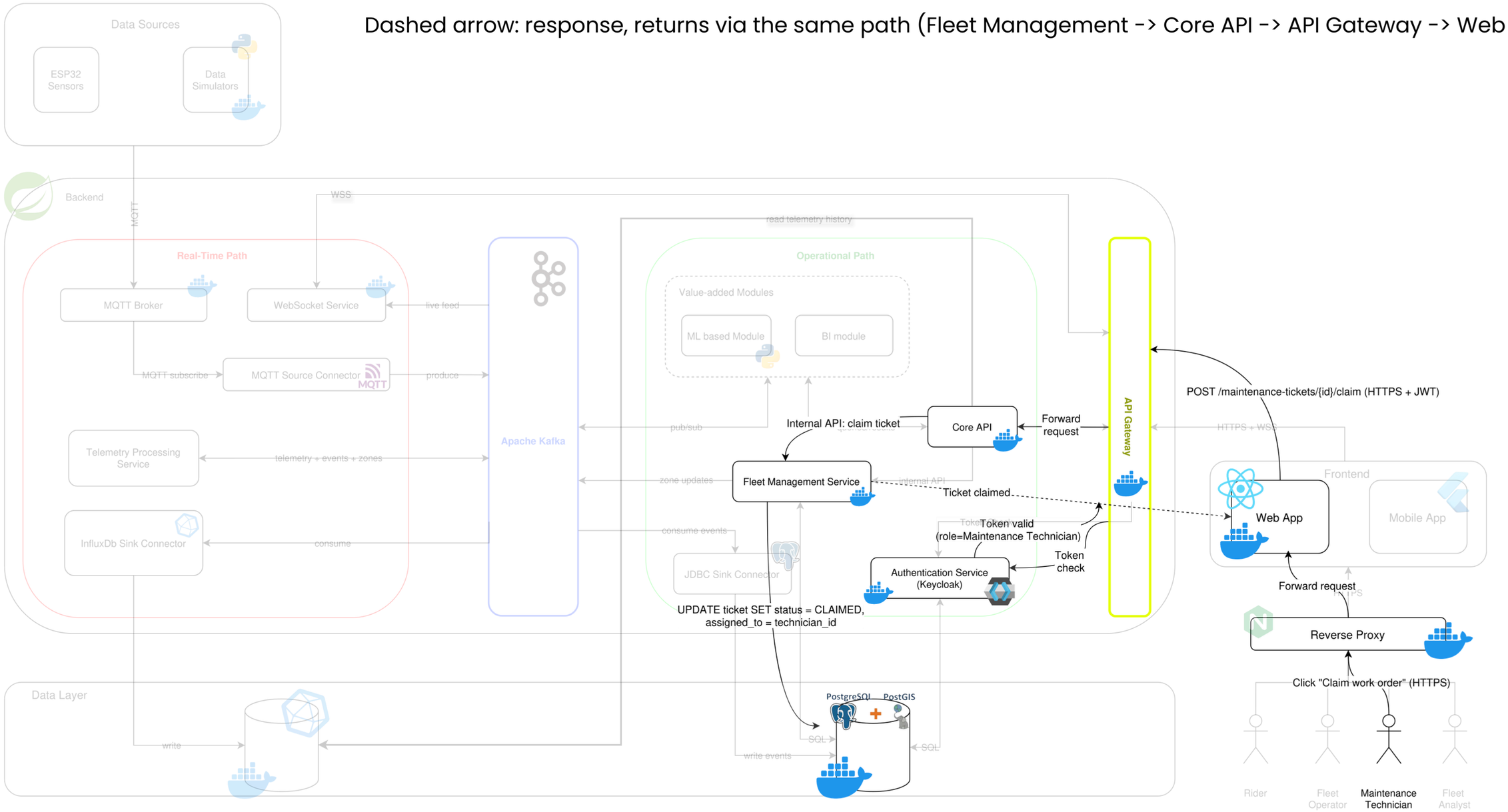
ARCHITECTURE WORKFLOW

Dashed arrow: response, returns via the same path (Fleet Management -> Core API -> API Gateway -> Web App)

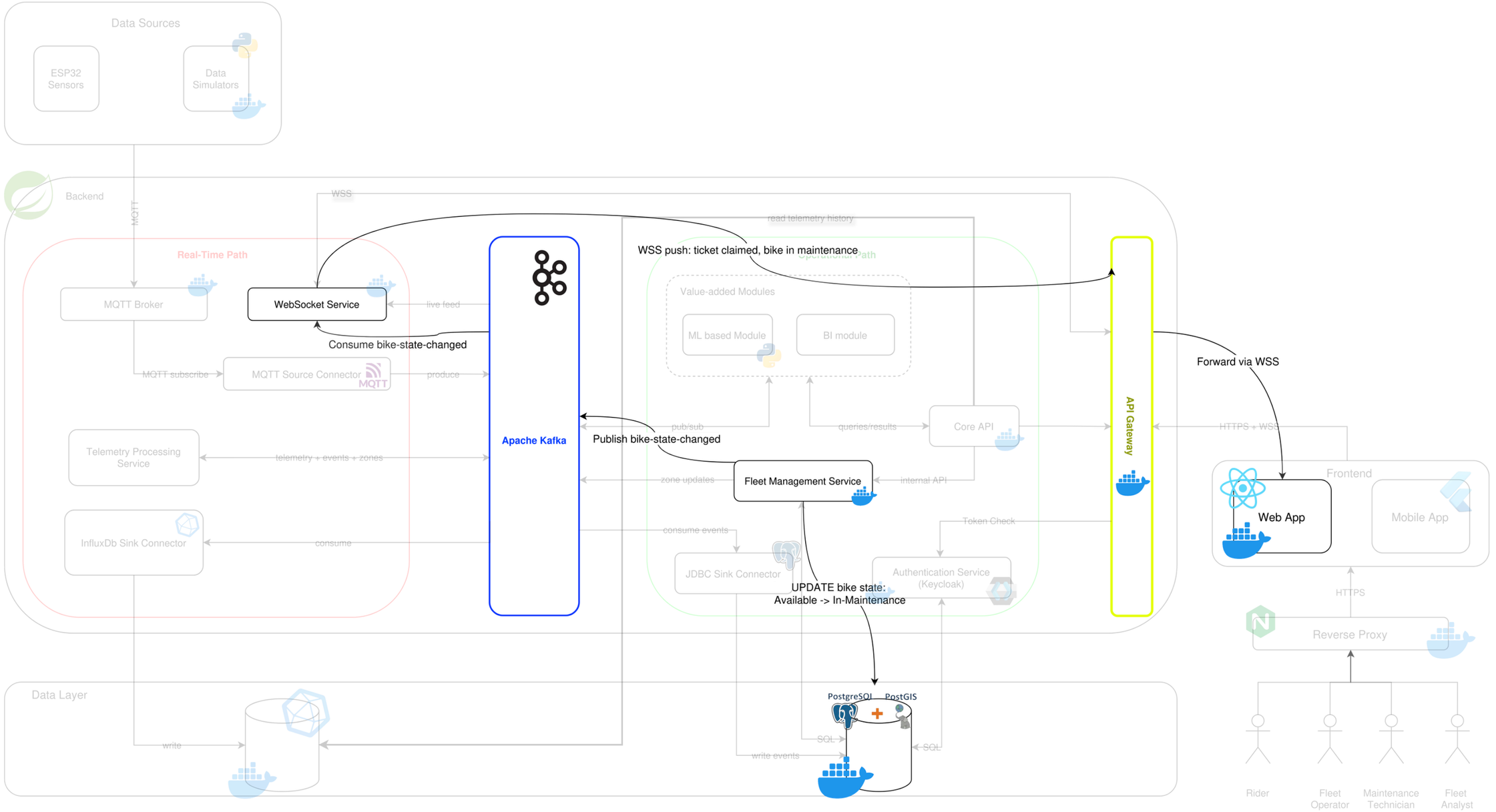


ARCHITECTURE WORKFLOW

Dashed arrow: response, returns via the same path (Fleet Management -> Core API -> API Gateway -> Web App)



ARCHITECTURE WORKFLOW



CALENDAR

Milestone	Date	Tasks
M2	30/09-06/10	7 - Requirements Gathering & Scope Definition (João Tomásio - Jira logging and management; Maria - initial functional requirements structuring) 8 - Actors, Personas & Use Cases (João Pereira and Guilherme - operator and user profile definition; João Tomásio - core use case mapping) 9 - Event-Driven Architecture Initial Design (Santiago - core infrastructure; Maria - telemetry data streams; João Pereira - UI integration touchpoint mapping)
	07/10-13/10	10 - Event-Driven Architecture Design & Kafka/MQTT Setup (Santiago - broker technical setup; Maria - data schemas and topic design) 11 - Non-Functional Requirements & Documentation (João Tomásio - constraints and security; João Pereira - usability requirements and UI/UX constraints) 12 - Presentation of Lifecycle Architecture (Guilherme - defense coordination and slide deck technical structuring with team-wide contributions)

CALENDAR

Milestone	Date	Tasks
M3	14/10-20/10	13 - Initial Mockups & Dashboard UI Concepts 14 - Main User Flows for Fleet Operators
	21/10-03/11	15 - Interactive Prototype & Design System 16 - Validation of Telemetry Data Ingestion
M4	04/11-24/11	Core Backbone & MVP: • Real-time Telemetry Ingestion Pipeline • Event Store & Database Setup • Individual Bike Registration & Lifecycle State • Aggregated Fleet Metrics API

CALENDAR

Milestone	Date	Tasks
	25/11-15/12	MVP Completion & Integration: • Service Layer: CO2 Footprint Tracking Module • Usage & Rider Analytics Module • Integration & Testing • Presentation
	16/12-22/12	1 - MVP Stabilization 2 - Review & Second-Semester Planning
	23/12-09/02	Christmas break & Exam period — code refactoring
	10/02-23/02	1 - Extensible API Framework & ML Module Integration (Demand Forecasting / Anomaly Detection)

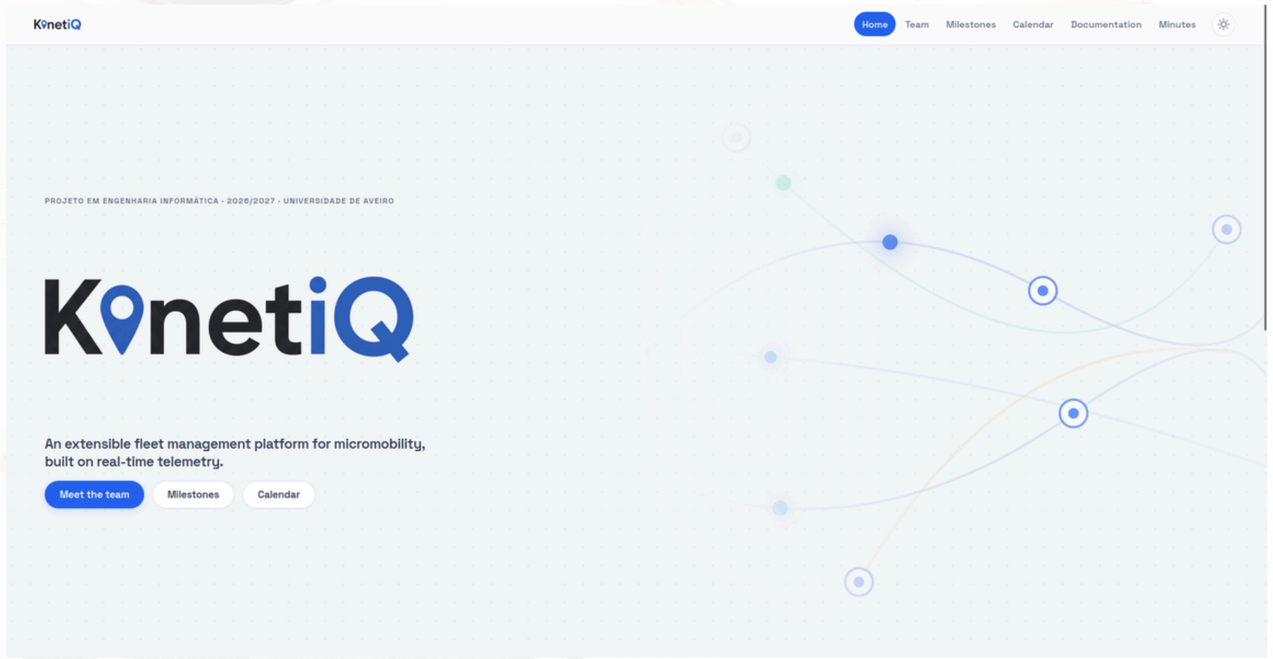
CALENDAR

Milestone	Date	Tasks
	24/02-09/03	2 - Operator Backoffice Dashboard UI (Telemetry & Service KPIs)
	10/03-23/03	3 - BI Reporting Module Integration
	24/03-02/04	Easter Holidays — code refactoring
	05/04-18/04	4 - Advanced Analytics & ML Model Refinement
	19/04-25/04	5 - Integration, Testing & Bug Fixing
	26/04-30/04	Academic Week — development suspension

CALENDAR

Milestone	Date	Tasks
	03/05-16/05	6 - User Testing 7 - Apply Improvements
	17/05-30/05	8 - Data Collection & Evaluation 9 - Stabilize Final Product
	31/05-03/06	10 - Create Documentation & Technical Report (Guilherme) 11 - Prepare Final Presentation and Defense
STUDENTS @DETI	04/06	• Demo • Poster • Video • Technical Report

THANK YOU



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